AERSP 309 — HW 3 formula sheet

$\mu_\oplus = 3.986 \times 10^5$ km^3/s^2, $R_\oplus = 6378$ km

Orbit geometry

$$r = \frac{p}{1 + e\cos\theta} \qquad p = \frac{h^2}{\mu} = a(1-e^2) = \frac{b^2}{a}$$

$$r_p = a(1-e), \quad r_a = a(1+e), \quad a = \tfrac{1}{2}(r_p + r_a)$$

$$e = \frac{r_a - r_p}{r_a + r_p}, \quad c = ae, \quad b = a\sqrt{1-e^2}$$

$$h = \sqrt{\mu p} = r\, v \cos\gamma = r_p v_p = r_a v_a$$

θ is the true anomaly (called f in the HW). Check any formula for p with a circle: $r_p = r_a = r$ must give $p = r$.

Energy, speed, period

$$\varepsilon = \frac{v^2}{2} - \frac{\mu}{r} = -\frac{\mu}{2a}$$

$$v^2 = \mu\left(\frac{2}{r} - \frac{1}{a}\right) \text{ (vis-viva)}$$

$$v_c = \sqrt{\frac{\mu}{r}}, \qquad v_{esc} = \sqrt{\frac{2\mu}{r}}$$

$$T = 2\pi\sqrt{\frac{a^3}{\mu}}, \qquad n = \frac{2\pi}{T} = \sqrt{\frac{\mu}{a^3}}$$

$\varepsilon < 0$: ellipse; $\varepsilon = 0$: parabola; $\varepsilon > 0$: hyperbola.

Velocity and flight-path angle

$$v_r = \frac{\mu}{h}\, e\sin\theta, \qquad v_\theta = \frac{\mu}{h}\,(1 + e\cos\theta)$$

$$\tan\gamma = \frac{v_r}{v_\theta} = \frac{e\sin\theta}{1 + e\cos\theta}$$

$$\cos\gamma = \frac{h}{rv}$$

$$\dot\theta = \frac{h}{r^2} = \sqrt{\frac{\mu}{p^3}}\,(1 + e\cos\theta)^2$$

Finding θ from r: $\cos\theta = \dfrac{p/r - 1}{e}$. Quadrant: $\gamma > 0$ (moving away from the planet) means $0 < \theta < 180°$; $\gamma < 0$ (approaching) means $180° < \theta < 360°$.

Time on the orbit (Kepler's equation)

$$M = E - e\sin E = n\,(t - t_p)$$

$$r = a\,(1 - e\cos E)$$

$$\cos E = \frac{e + \cos\theta}{1 + e\cos\theta}$$

$$\Delta t = \frac{M_2 - M_1}{n}$$

Time is proportional to swept area, not to angle. E lies in the same half-plane as θ (both in 0–180° or both in 180°–360°). For time spent over a symmetric arc about periapsis, double the one-sided time.

Parabola ($e = 1$) and hyperbola ($e > 1$)

Parabola

$$v = \sqrt{\frac{2\mu}{r}}, \qquad r_p = \frac{p}{2}, \qquad \gamma = \frac{\theta}{2}$$

$$v_\infty = \sqrt{-\frac{\mu}{a}} = \sqrt{2\varepsilon}$$

Hyperbola ($a < 0$)

$$r_p = a(1-e), \qquad p = a(1-e^2)$$

$$\theta_\infty = \cos^{-1}\left(-\frac{1}{e}\right), \qquad \delta = 2\sin^{-1}\frac{1}{e} = 2\theta_\infty - 180°$$

Problem 1: velocity frame $\mathcal{V}$ vs. orbit frame $\mathcal{O}$

$$\tan\beta = \frac{e\sin f}{1 + e\cos f}, \quad \sin\beta = \frac{e\sin f}{k}, \quad \cos\beta = \frac{1 + e\cos f}{k}, \quad k = \sqrt{1 + 2e\cos f + e^2}$$

$\underbrace{C^{\mathcal{VO}}}_{\text{rows: } \mathcal{V} \text{ axes in } \mathcal{O}}$ is a single rotation about $\hat\imath_h$ by $\pm\beta$ (get the sign from the figure)

$$\vec\omega^{\mathcal{V}/\mathcal{P}} = \vec\omega^{\mathcal{V}/\mathcal{O}} + \vec\omega^{\mathcal{O}/\mathcal{P}}, \qquad \vec\omega^{\mathcal{O}/\mathcal{P}} = \dot f\,\hat\imath_h, \qquad \dot f = \sqrt{\frac{\mu}{p^3}}\,(1 + e\cos f)^2$$

$$\vec{\alpha} = \frac{d\omega}{dt}\,\hat{\imath}_h = \dot{f}\,\frac{d\omega}{df}\,\hat{\imath}_h \quad (\hat{\imath}_h \text{ is fixed in both frames}); \qquad \dot{\beta} = \dot{f}\,\frac{d\beta}{df} \text{ from differentiating } \tan\beta$$

Problem 4: orbital elements from $\vec{r}_1, \vec{r}_2$

$$\hat{h} = \frac{\vec{r}_1 \times \vec{r}_2}{|\vec{r}_1 \times \vec{r}_2|}, \qquad \cos i = \hat{h}_3, \qquad \vec{n} = \hat{n}_3 \times \hat{h} = (-\hat{h}_2,\ \hat{h}_1,\ 0), \qquad \Omega = \operatorname{atan2}(n_2, n_1)$$

$$\cos\Delta f = \frac{\vec{r}_1 \cdot \vec{r}_2}{r_1 r_2}, \qquad f_2 = f_1 + \Delta f \qquad (\text{assumes the short way, } \Delta f < 180^\circ)$$

$$e\cos f_1 = \frac{p}{r_1} - 1, \qquad e\cos(f_1 + \Delta f) = \frac{p}{r_2} - 1 \implies \text{expand to get } e\sin f_1$$

$$e = \sqrt{(e\cos f_1)^2 + (e\sin f_1)^2}, \quad f_1 = \operatorname{atan2}(e\sin f_1,\ e\cos f_1), \quad a = \frac{p}{1-e^2}$$

$$\cos u = \hat{n}\cdot\hat{r} \quad (u \to 360^\circ - u \text{ if } r_3 < 0), \qquad \omega = u - f$$

Ascending vs. descending: sign of γ (i.e. of $\sin f$) says moving away from or toward the planet; sign of $\dot{r}_3$ says north or south. State which you mean.

Problem 3 (proofs) and textbook problems

P3: start from $r_p = a(1-e)$, $r_a = a(1+e)$, $b = a\sqrt{1-e^2}$, $p = b^2/a$. **2.16:** $\varepsilon \to a$, $h \to p \to e$, compare r_p with $R_\oplus$. **2.17:** same r_p; T_B/T_A from the angle A must cover; $a \propto T^{2/3}$; $r_a = 2a - r_p$. **2.19, 2.20:** parabola formulas. **2.21:** $\gamma = 0$ means perigee; h, p, e; vis-viva; $\cos\gamma = h/rv$; $v_\infty = \sqrt{2\varepsilon}$. **2.22:** $r_p = a(1-e)$, $p = a(1-e^2)$, v_r, v_θ, θ_∞, δ. **2.28:** a, e from r_p, r_a, then T. **2.32:** a from ε, p from h, then e; use the Moon's μ and radius. **2.42:** v_∞ from a; $r_p = a(1-e)$; use Jupiter's μ.